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RESEARCH PAPER

A Relevant Antecedent of Flow Experience: Task Meaningfulness

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Abstract
Flow experience is often labeled as optimal experience or being in the zone. One research goal is to better understand the antecedents of flow to ultimately identify ways to increase it in interventions. We present a theoretical approach that considers task meaningfulness—the extent to which a certain task is perceived as personally meaningful—as a crucial antecedent of flow experience. We tested this proposition in a controlled experimental setting by having participants play Tetris and in everyday life by conducting a five-day ambulatory assessment study (both studies were preregistered; total $N = 1{,}062$). In the experimental setting, task meaningfulness had a causal and positive impact on flow experience. In everyday life, both constructs were positively associated across various situational contexts, even when controlling for skills-demands fit—the most established antecedent of flow. Our findings suggested that task meaningfulness and skills-demands fit were differentially associated with the experiential dimensions of flow. Task meaningfulness was more strongly related to the flow dimensions of absorption and intrinsic reward, whereas skill-demands fit showed a closer association with effort-less control. Crucially, our findings consistently demonstrated that task meaningfulness and the intrinsic reward dimension of flow constitute distinct constructs, thereby mitigating concerns of circular reasoning in our theoretical rationale, empirical evidence, and conclusions. Overall, we provide internally and externally valid evidence that task meaningfulness is a relevant yet previously overlooked antecedent of flow, offering a foundation for innovative and effective flow interventions.

Keywords Flow experience · Task meaningfulness · Motivation · Ambulatory assessment · Everyday life

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1 Introduction

In this paper, we examine the previously untested role of the antecedent task meaningfulness for the experience of flow. We aimed to test whether task meaningfulness is an incrementally valid antecedent of flow beyond the 'classic' antecedent skills-demands fit. The present research may help to develop a new starting point for modulating people's flow experiences (for an overview, see Bartholomeyczik et al., 2023). In the following, we outline the theoretical background for this proposition and present experimental and ambulatory assessment data examining the relation between the two constructs both in a controlled and in an everyday life setting.

2 Flow Experience and its Antecedents

Although early studies examined flow during peak performance by chess players, surgeons, dancers, and mountain climbers (Csikszentmihalyi, 1975), we now know that 'ordinary people' can have optimal experiences or be in the zone during work, during leisure time (Bartholomeyczik et al., 2024; Engeser & Baumann, 2016), and even under laboratory conditions when playing simple computer games (Baumann et al., 2016; Keller & Bless, 2008; Keller & Blomann, 2008; Scheepers & Keller, 2022; Zhang et al., 2023) or working on mental arithmetic tasks (Peifer et al., 2020; Ulrich et al., 2014). In his early work on the flow experience, Csíkszentmihályi (1975) proposed the so-called channel model based on the observation that individuals are most likely to enter the flow state when the demands inherent in the activity are perceived to fit a person's skills. This assumption has been tested by various research groups in different contexts and can be considered empirically well-supported (Baumann et al., 2016; Engeser & Rheinberg, 2008; Harmat et al., 2015; Keller & Bless, 2008; Keller & Blomann, 2008; see Melnikoff et al., 2022, for a contrary view). In addition to the consistent empirical support for its validity, the flow channel model has desirable properties from a philosophy of science perspective (Glöckner et al., 2018). First, it is a parsimonious model of human motivation that is based on a simple IF component: skills-demands fit. Second, the THEN component is very precise in that it describes a specific state of experience. In this work, we apply the perspective proposed by Norsworthy and colleagues (2021, 2023), who suggested that flow is characterized by the experience of absorption, effort-less control, and intrinsic motivation. Sample items from the respective Psychological Flow Scale include, 'All my attention was on the task/activity,' 'There was a sense of fluidity to my actions,' and 'I would like the feeling of that experience again.' Thus, the flow channel model can be considered as a sound theoretical framework. There is evidence, however, that skills-demands fit is not the only construct relevant to the emergence of flow. Work examining the role of personality traits found that action-orientation, that is, the tendency to stay engaged and focused until a task is completed (Diefendorff et al., 2000), moderated the relation between skills-demands fit and flow. Specifically, high (vs. low) levels of action-orientation were associated with intense flow even under perceived non-fit (Baumann et al., 2016; Keller & Bless, 2008). This suggests that the flow channel model is overly parsimonious and requires one or more complementary factors representing antecedents and/or boundary conditions of flow.

One line of research has expanded the flow channel model by examining personality traits that are beneficial or detrimental to flow (e.g., Baumann et al., 2016; Ullén et al., 2012). What has received less attention, in our view, is whether there are other characteristics of the activities that modulate the experience of flow. In the following sections, we outline our proposition that task meaningfulness should be considered as a relevant antecedent for the emergence of flow.

3 Definition of the Construct Task Meaningfulness

Although the work psychology literature shows that there is no clear distinction between the causes of meaningful work and the experience of meaningful work itself, as Steger et al. (2012) noted, the empirical evidence documenting the significance of meaningful work is still impressive. Employees who find their work meaningful report higher levels of well-being (Arnold et al., 2007), experience greater job satisfaction, and report higher levels of life satisfaction (Steger et al., 2012). Steger et al. (2012) defined the concept as follows:

We define [meaningful work] not as simply whatever work means to people (meaning), but as work that is both significant and positive in valence (meaningfulness). Furthermore, we add that the positive valence of [meaningful work] has a eudaimonic (growth- and purpose-oriented) rather than hedonic (pleasure-oriented) focus. (p. 322)

Specifically, Steger et al. conceptualized meaningful work as a multidimensional construct, consisting of (a) the subjective perception that work is meaningful, (b) meaning-making through work (i.e., work as a source of meaning in life), and (c) greater goods motivation (i.e., work has a positive impact on others). The three dimensions are strongly correlated and the first one is 'the "flagship" indicator of the overall construct of meaningful work' (Steger et al., 2012, p. 333) because it is most strongly associated with well-being. We refer to this core definition and conceptualize meaningful work as the subjective experience that what one does at work is meaningful (see also Hackman & Oldham, 1976; Rosso et al., 2010). Because we want to understand the role of the construct meaningfulness on a general level, that is, beyond the context of work, we apply the following conceptualization of task meaningfulness in relation to any given activity: a subjective experience that a particular task or activity is meaningful (see also Bailey & Madden, 2016). Thus, we view meaningful work as a subset of the more general construct task meaningfulness.

4 Theoretical Embedding of the Construct Task Meaningfulness

Although task meaningfulness has not been critically examined in the context of flow, the concept has been considered in other motivational theories. One prominent example is the Job Characteristics Model (JCM; Hackman & Oldham, 1976), which posits that certain job characteristics, such as task significance, have an impact on work-related outcomes (e.g., motivation and satisfaction) that are mediated by psychological states such as meaningfulness. The definition of meaningfulness in the JCM is similar to the definition of meaningful work (Allan et al., 2018; Steger et al., 2012); thus, the terms can be considered synonymous. Referring to the JCM not only helps to provide clear conceptualization of the main construct for the present work, but it also provides evidence that meaningfulness may be related to

flow. Meaningfulness (in the sense of meaningful work) has been positively associated with criteria such as job performance (Allan et al., 2018; Hackman & Oldman, 1976; Humphrey et al., 2007), job satisfaction (Fried & Ferris, 1987; Humphrey et al., 2007), life satisfaction (Arnold et al., 2007; Steger et al., 2012), intrinsic work motivation (Fried & Ferris, 1987; Humphrey et al., 2007; Steger et al., 2012), and work engagement (May et al., 2004). The last two findings are particularly noteworthy because of their clear conceptual overlap with flow. That is, the positive associations of meaningfulness with work engagement and intrinsic work motivation support the notion that task meaningfulness is a candidate to emerge as antecedent of flow.

Self-Determination Theory (SDT; Ryan & Deci, 2017) posits that self-relevance[1]—a construct closely related to task meaningfulness—has a causal impact on motivation (Vansteenkiste et al., 2012). According to SDT, motivation can be broadly categorized into two types: intrinsic and extrinsic motivation. Intrinsically motivated activities are those that individuals find inherently interesting or enjoyable. These activities possess the highest degree of self-relevance and do not require internalization, as the motivation is fully autonomous and stems directly from the activity itself (Ryan & Deci, 2000, 2017). In contrast, extrinsically motivated activities are performed as a means to an end—that is, to attain outcomes that are separate from the activity itself. Such outcomes may vary widely, and the underlying reasons for engaging in extrinsically motivated behavior differ in the extent to which they are perceived as self-relevant and are internalized. SDT emphasizes that extrinsic motivation is not a unitary construct; rather, it comprises multiple forms that vary in their degree of self-relevance, autonomy and internalization (Ryan & Deci, 2000). For instance, external regulation represents the least autonomous form of extrinsic motivation, where behavior is driven by external rewards or threats of punishment. The activity itself is typically low in appeal and perceived self-relevance. Conversely, identified regulation represents a more autonomous form, wherein individuals perceive the activity as self-relevant and have internalized its value (Ryan & Deci, 2000). For example, someone may engage in physical exercise—not because it is inherently enjoyable—but because they value its benefits, such as maintaining health, socializing with friends, or managing stress. The role of self-relevance as a causal factor in shaping autonomous motivation via internalization has received considerable attention in educational psychology (Vansteenkiste et al., 2018). For example, Reeve et al. (2002) found that students who were given a rationale for why the learning material might be personally meaningful reported greater classroom engagement compared to those who were not. Similarly, Jang (2008) demonstrated that instructional practices that support autonomy by highlighting personal relevance can significantly enhance student motivation and effort. A meta-analysis suggests that a provision of a rationale not only positively affects engagement, but shows beneficial effects on performance as well (Steingut et al., 2017). These findings, along with the theoretical considerations from SDT, suggest that task meaningfulness may represent a distinct antecedent of flow.

Although we find the argument that task meaningfulness and flow are distinct constructs convincing, one could also adopt a different and more critical perspective. For instance, it could be argued that in intrinsically motivated activities, the experience of flow and per-

[1] There is no consistent use of this term in the SDT literature. The following terms can be understood as synonyms: ‚personal meaning' (Vansteenkiste et al., 2018); ‚subjective task value' (Steingut et al., 2017); ‚personal value' (Reeve et al., 2002); ‚task value' (Jang, 2008); and ‚perceived importance/task importance' (Jang, 2008; Reeve et al., 2002).

ceived task meaningfulness are so closely linked that they cannot be separated into antecedent and outcome. A similar point has been raised in the field of work psychology, where it has been acknowledged that the reasons for meaningful work and its experience are sometimes difficult to distinguish (Steger et al., 2012). If this critical view were correct, the present work would risk being based on a circular argument. To address this concern, we examined whether the constructs of task meaningfulness and the flow dimension of intrinsic reward can be empirically differentiated.

5 Comparable Ideas from the Literature on Flow Experience

The idea that the characteristics of the activity may influence flow has been discussed by three different research groups but has not yet been empirically tested. First, Tozman and Peifer (2016) suggested considering the self-relevance of an activity to resolve conflicting findings regarding the physiological correlates of flow (see also Peifer & Wolters, 2017). The construct of self-relevance overlaps strongly with the conceptualization of task meaningfulness as defined above. The term used by Tozman and Peifer is rooted in the biopsychosocial model of challenge and threat (Blascovich & Tomaka, 1996), which is a motivational theory that applies to skill-related tasks that are perceived as goal-relevant. Blascovich et al. (2003) defined goal-relevance as 'the extent to which individuals perceive the task as having meaning for the self' (p. 238), which implies strong parallels to task meaningfulness. Second, Landhäußer and Keller (2012) suggested investigating the subjective value of an activity when searching for additional factors that influence the flow experience (see also Barthelmäs & Keller, 2021). The term comes from the regulatory fit literature (Higgins, 2006) and is defined as 'a motivational force of attraction to or repulsion from something' (Higgins & Scholer, 2009, p. 100). As such, the construct has crucial similarities to the conceptualization of task meaningfulness. Third, in a recent scoping review, Norsworthy et al. (2021) identified high motivational force as a potential antecedent of flow, including constructs such as interest, importance, subjective value, and goal congruency and stated that 'further research is needed to better understand the specific types of motivation that are most conducive to flow experiences and how they relate, or not, to "optimal challenge"' (p. 815). Our work can be seen as a direct response to this call. In summary, ideas similar to those underlying the present work have been put forward in the existing flow literature, but none have been empirically tested.

6 The Present Research

With this research, we aimed to examine whether task meaningfulness is a relevant antecedent of flow experience. In a controlled experimental setting using a Tetris game (Study 1), we tested whether task meaningfulness causally influenced the experience of flow and expected higher flow experience in the meaningfulness condition compared to the control condition. To complement this with externally valid data, we also investigated the association between task meaningfulness and flow in everyday life through an ambulatory assessment study (Study 2) and expected a positive association between both constructs. In both studies, we also explored whether the constructs of task meaningfulness and the flow dimen-

sion of intrinsic reward could be empirically distinguished, in order to guard against circular reasoning in our theoretical considerations, empirical findings, and conclusions.

7 Study 1

7.1 Methods

7.1.1 Participants

Participants were recruited via Prolific. All participants agreed to an informed consent and received £1.40 (approx. €1.70) as compensation. The study was approved by the local ethics committee and was preregistered under: https://aspredicted.org/rzr8-vyy5.pdf. All participa nts were from Germany, and they were M_{age} = 33.9 (SD_{age} = 11.6; $range_{age}$ = 18–65) years old. From the N=513 participants, 212 identified as women, 296 as men, 4 as nonbinary/genderqueer, and 1 did not want to specify. An a priori power analysis with G*Power (Faul et al., 2007) suggested that a sample size of N=506 was sufficient to detect a group difference of d=0.25 (obtained from a pilot study) with 80% power.

7.2 Procedure

Participants were randomly assigned to one of two conditions. In the meaningfulness condition (n=249), they were informed that for every 20 points earned in the Tetris game (this corresponds to two completed lines), one euro cent would be donated to a charity of their choice (all donations were actually paid). They could select from eight different charities and were then asked to explain their choice in 3–5 sentences (see supplemental material in the OSF for details, Table S1, https://osf.io/d7jrs). In the control condition, Tetris perform ance was not linked to any donations (n=264). This manipulation of task meaningfulness is comparable to the approach of Westgate and Wilson (2018; Study 2). Participants were first introduced to the rules of Tetris and given a 60-second trial to familiarize themselves with the controls. After this practice phase, we assessed skills-demands fit and task meaningfulness—ensuring that these evaluations were not yet influenced by the experience of flow. Next, the rules were reiterated, and participants in the meaningfulness condition were reminded that their performance was linked to a charity donation. Participants then played Tetris for six minutes (comparable to Scheepers and Keller, 2022), after which we measured their flow experience. The study contained an attention-check item, and participants who answered it incorrectly were excluded from the study.

7.3 Tetris Game

The goal of the Tetris game is to arrange falling objects so that they form fully complete lines at the bottom of the playing field. Players can move the objects left or right, rotate them in 90-degree increments, and accelerate their descent using designated keyboard keys. The game lasted six minutes, with the speed dynamically adjusted based on player performance. Specifically, if a player successfully completed five or more lines within a maximum of 30 consecutive pieces, the speed increased by one level. Conversely, if the player completed

three or fewer lines within the same limit, the speed decreased by one level (Keller & Bless, 2008; Keller & Blomann, 2008; Scheepers & Keller, 2022). Performance was defined as the total number of completed lines during the game. For each completed line, participants received 10 points. The game was continuous, meaning there was no "Game Over" when an object reached the top of the field. Instead, the playfield reset, removing all visible objects while preserving the player's score, allowing them to continue increasing their points.

7.4 Measurements

All responses were given on 7-point Likert scales ranging from 1 (*strongly disagree*) to 7 (*strongly agree*).

7.4.1 Flow Experience

We measured flow experience with the Psychological Flow Scale (Norsworthy et al., 2023), which consists of nine items. Three items each can be summarized into three dimensions, namely an absorption (sample item: 'I was absorbed in the act/task'; $\omega = 0.88$), an effort-less control (sample item: 'My actions flowed effortlessly'; $\omega = 0.83$), and an intrinsic reward (sample item: 'I found the experience rewarding'; $\omega = 0.87$) dimension, with all items averaged into a global flow score ($\omega = 0.89$). Because there is no German version of the scale, the first and second authors translated the items into German separately and reviewed the results together with the last author. Each item was discussed (and further adjusted) until agreement was reached. We followed the same procedure for all scales without German versions. The final version of all translations can be found in the supplemental material (Tables S2, S3, and S4). Participants were explicitly instructed to respond to the flow items in relation to the six-minute Tetris game.

7.4.2 Antecedents of Flow

Skills-demands fit was measured with the item 'The challenge and my skills were at an equally high level' from the Flow State Scale (Jackson & Marsh, 1996). We measured task meaningfulness with three items from the meaningfulness scale by May et al. (2004) and adapted the content of the May et al. items by removing the specific reference to work and relating the items to an activity. We also shortened the scale to three items. The items read, 'The activity is very important to me,' 'The activity is personally meaningful to me,' and 'The activity is personally valuable to me' ($\omega = 0.93$).

7.4.3 Additional Measures

Additional measures and analyses are provided in the supplemental material (Table S5, Table S6). Briefly, we included an additional item each for underload and overload to assess skills-demands fit in greater detail. Two extra flow-related items were added to capture altered time perception and reduced self-awareness. For the Tetris game, we recorded the number of playfield resets, level increases, and level decreases.

7.5 Data Analysis

We used a Welch's t-test to test our main hypothesis, that is, we compared flow experience between the meaningfulness and the control condition. Another Welch's t-test was performed to assess the effectiveness of the manipulation (manipulation check). The following exploratory analyses were conducted. We explored whether and how the manipulation affected the flow dimensions, skills-demands fit and performance. Further, we examined the associations between skills-demands fit, task meaningfulness and flow experience with correlations and regressions. Using confirmatory factor analyses (CFA), we explored whether task meaningfulness and the intrinsic reward dimension of flow could be empirically distinguished. To do so, we specified the measurement model for flow such that three items loaded onto each of the three flow dimensions—absorption, effort-less control, and intrinsic reward—which in turn loaded onto a higher-order global flow factor. We modeled the three task meaningfulness items in two alternative ways: either loading onto a separate task meaningfulness factor (at the same level as the flow dimensions and not loading onto the global flow factor), or loading directly onto the intrinsic reward dimension. Analyses were conducted using R (R Core Team, 2024; version 4.4.1) within RStudio (Posit team, 2024, 2024.4.2.764) and the following packages: lavaan (Rosseel, 2012) and semTools (Jorgensen et al., 2022). Reliability estimates were obtained as suggested by Flora (2020).

7.6 Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. Data, analysis code, and research materials are available at https://osf.io/d7jrs. This study's design and its analysis were preregistered under: https://aspredicted.org/rzr8-vyy5.pdf.

7.7 Results

The measurement model for flow and task meaningfulness yielded adequate model fit (CFI = 0.98, TLI = 0.97, RMSEA = 0.06, SRMR = 0.05). The loadings of the items on the three flow dimensions (and the loadings on the global flow factor) as well as the task meaningfulness factor were positive and significant (see supplemental material). Letting the task meaningfulness items load on the intrinsic reward dimension of flow (i.e., six items for this dimension) resulted in a poor model fit (CFI = 0.78, TLI = 0.71, RMSEA = 0.19, SRMR = 0.18), indicating that task meaningfulness and the intrinsic reward dimension of flow are distinct constructs. The correlations of the separate items are depicted in Table S7 (supplemental material), further supporting this reasoning.

Table 1 contains descriptive statistics and the group comparisons for flow experience, task meaningfulness, skills-demands fit and performance (additional analyses are depicted in the supplemental material). Flow experience and task meaningfulness were higher in the meaningfulness compared to the control condition. On closer inspection, group differences were most pronounced for the subdimensions intrinsic reward ($d = 0.28$), and absorption ($d = 0.19$), compared to effort-less control ($d = 0.07$; Table S8). Skills-demands fit and performance in the Tetris game did not differ significantly between the conditions. The correlation between flow and skills-demands fit was $r = 0.29$, the correlation between flow and

Table 1 Descriptive statistics and group comparisons for study 1

	Flow Experience	Task Meaningfulness	Skills-Demands Fit	Performance[a]
M (SD) per Condition				
Meaningfulness	5.24 (0.99)	3.71 (1.53)	4.36 (1.59)	292.77 (158.01)
Control	5.01 (1.02)	3.21 (1.52)	4.16 (1.52)	280.57 (150.36)
Group Comparison	$d=0.22$ $t(510.44)=2.52$, $p=0.012$	$d=0.33$ $t(509.02)=3.71$, $p<0.001$	$d=0.13$ $t(505.75)=1.42$, $p=0.157$	$d=0.08$ $t(505.10)=0.89$, $p=0.371$

[a] for each completed line, participants received 10 points. The highest donation amount was 40 euro cents, corresponding to one participant who achieved a score of 800 points

task meaningfulness was $r=0.39$ (averaged across both conditions; Table S9). Regression analyses revealed that task meaningfulness and skills-demands fit incrementally predicted flow ($R^2_{\text{skills–demands fit}} = 0.08$, delta $R^2_{\text{task meaningfulness}} = 0.11$), the interaction term of both predictors was not significant (Tables S10, S11, and S12).

7.8 Discussion

Study 1 provides experimental evidence that task meaningfulness has a causal and positive impact on the experience of flow, supporting our main hypothesis. At the level of flow dimensions, the effect was particularly pronounced for intrinsic reward and absorption compared to effort-less control. This suggests that task meaningfulness is especially relevant for these specific dimensions of flow experience. Skills-demands fit and task meaningfulness were positively associated with flow experience, further supporting our main proposition. While the impact of the manipulation on skills-demands fit was small and not significant, a preliminary interpretation could be that task meaningfulness and skills-demands fit are largely, but not entirely, independent antecedents of flow. This interpretation is also supported by the regression analyses, which indicated that both antecedents are incrementally valid, while no statistically significant interaction was found.

We used well-established methods to examine flow experience in a controlled setting (e.g., Keller & Bless, 2008) and to manipulate task meaningfulness (Westgate & Wilson, 2018, Study 2). Furthermore, the manipulation check confirmed that the manipulation had the intended effect. Skills-demands fit and task meaningfulness were assessed after a short practice phase (i.e., before the six-minute Tetris game), ensuring that their evaluation was not influenced by the flow experience. The CFA results demonstrated that task meaningfulness and the flow dimension of intrinsic reward are empirically distinct constructs, reducing the risk of circular reasoning in our theoretical considerations, empirical findings, and conclusions. In sum, Study 1 provides internally valid results. However, the specific context of the Tetris game raises concerns about the generalizability of the findings to other contexts. Following the perspective that psychological research should place greater emphasis on the generalizability and external validity of findings (Diener et al., 2022), we conducted an ambulatory assessment study to examine whether task meaningfulness is associated with flow in everyday life. Furthermore, we tested whether this association remained robust when controlling for skills-demands fit during the episode, which is considered the most well-established antecedent of flow. In addition, we assessed individual differences—measured at baseline through basic personality traits from the HEXACO model—to ensure that

Table 2 Descriptive Statistics, bivariate correlation coefficients and intraclass correlation coefficients

Variable	1	2	3	4	5	6	7	8	9	10	11	12
1. Flow		0.90	0.87	0.89	0.76	0.64	0.14	0.03	0.34	0.13	0.19	0.00
2. Flow-absorption	0.76		0.67	0.71	0.66	0.54	0.14	0.05	0.25	0.10	0.19	−0.01
3. Flow-effort-less control	0.76	0.34		0.65	0.60	0.64	0.20	−0.05	0.32	0.13	0.18	−0.08
4. Flow-intrinsic reward	0.86	0.46	0.55		0.76	0.52	0.02	0.09	0.33	0.11	0.13	0.08
5. Task Meaningfulness	0.58	0.42	0.33	0.60		0.56	0.15	0.09	0.28	0.08	0.20	0.01
6. Skills-Demands Fit	0.34	0.26	0.33	0.25	0.27		0.11	0.04	0.26	0.08	0.08	0.00
7. Honesty-Humility								−0.04	0.04	0.21	0.24	0.05
8. Emotionality									−0.24	−0.06	0.13	0.03
9. Extraversion										0.12	0.12	0.19
10. Agreeableness											0.06	0.02
11. Conscientiousness												0.10
12. Openness												
M	5.00	4.92	5.32	4.75	5.13	5.27	4.78	4.31	4.34	4.41	4.99	4.48
SD within	0.94	1.21	1.00	1.32	1.31	1.26						
SD between	0.62	0.71	0.69	0.67	0.75	1.03	0.95	0.89	0.89	0.77	0.79	0.96
ICC	0.298	0.255	0.325	0.205	0.246	0.402						

Note. Correlation coefficients at the within-person level in the lower triangle, and correlation coefficients at the between-person level in the upper triangle; ICC = Intraclass Correlation Coefficient 1, i.e., proportion of between person variance to the total variance

the association between task meaningfulness and flow demonstrated incremental validity beyond these basic individual differences (e.g., Altgassen et al., 2024).

8 Study 2

8.1 Methods

8.1.1 Participants

Participants were recruited at the university campus via mailing lists and the market research polling company Bilendi. All participants agreed to an informed consent and received course credit or cash (approx. €15) as compensation. Specific compensation was based on the number of episodic measures completed, with a maximum of €20. The present data were collected as part of a larger ambulatory assessment project on motivational experiences in everyday life. This means that we do not report on all of the variables collected in the project, but only on those that are relevant to the research question at hand. The study was approved by the local ethics committee and was preregistered under https://aspredicted.org/bghk-3b92.pdf.

Sixteen participants were excluded due to failed attention checks. The baseline questionnaire was successfully completed by 680 participants and lasted approx. 30 min. Each episodic measure took approx. 5 min. As preregistered, we focused our analyses on participants with 10 or more episodic measures, therefore our final sample comprised $N = 549$ participants with $n = 10{,}232$ episodic measures. Participants provided $M = 18.6$ episodic measures on average (74% compliance), and were $M_{age} = 35.1$ ($SD_{age} = 13.6$; $\text{range}_{\text{age}} = 18–74$[2]) years old. Three hundred thirty-six participants identified as women, 208 as men, 2 as nonbinary/genderqueer, 1 as no gender, and 2 did not want to specify (detailed demographic information is presented in the supplemental material). An a priori power analysis following the approach by Lafit et al. (2021; model 3) and referring to pilot data suggested that a sample size of $N = 40$ was sufficient to detect the assumed association between task meaningfulness and flow with 100% power, hence, the study was well-powered.

8.2 Procedure

We used the software Telegram-Survey-Bot to conduct the ambulatory assessment project (Barthelmäs et al., 2021). Participants used their private smartphone (iOS or Android) for answering the self-report measures. At the start, participants were instructed to install the messenger app Telegram and to add a study-specific chatbot account to their contact list. Participants first completed a longer baseline survey and then worked on the ambulatory assessment phase for the next five consecutive days. They received five episodic measures each day, starting 60 min after waking up, with 190 min between measures. A respondent who got up at 7 a.m. was asked to complete an episodic measurement at 8 a.m., 11:10 a.m., 2:20 p.m., 5:30 p.m., and 8:40 p.m. To prevent participants from accurately predicting the episodic measures, the actual notification varied by 20 min around the calculated time. Participants were given detailed instructions for the episodic measures at baseline, which were

[2] Due to a programming error, age was not recorded for $n = 37$ participants.

summarized in each episodic measure. Participants were instructed to relate their responses to the activity they were engaged in just before the notification. In addition, the activity should have taken at least several minutes (e.g., writing an email) rather than a short action (e.g., opening a door). After each notification, the survey link was accessible for 45 min. This procedure should ensure that as few flow experiences as possible were interrupted by the episodic measures.

8.3 Measurements

HEXACO was measured at baseline. For each episodic measure, flow, antecedents of flow, and type of situation were assessed. Unless otherwise stated, responses were given on 7-point Likert scales ranging from 1 (*strongly disagree*) to 7 (*strongly agree*).

8.3.1 HEXACO-60 Inventory

The German version of the HEXACO-60 inventory (Moshagen et al., 2014) was used to assess the six HEXACO personality factors (i.e., 10 items for each of the six scales). Reliability estimates for all subscales were good (Honest-humility: $\omega = 0.77$; Emotionality: $\omega = 0.79$; Extraversion: $\omega = 0.79$; Agreeableness: $\omega = 0.75$; Conscientiousness: $\omega = 0.77$; Openness: $\omega = 0.77$).

8.3.2 Flow

We measured flow as described in Study 1. Reliability estimates were good (absorption: $\omega_{within} = 0.87$, $\omega_{between} = 0.84$; effort-less control: $\omega_{within} = 0.73$, $\omega_{between} = 0.82$; intrinsic reward: $\omega_{within} = 0.87$, $\omega_{between} = 0.76$; global flow score: $\omega_{within} = 0.87$, $\omega_{between} = 0.80$).

8.3.3 Antecedents of Flow

The antecedents of flow were measured as described in Study 1 (reliability estimates for task meaningfulness: $\omega_{within} = 0.91$, $\omega_{between} = 0.84$).

8.3.4 Type of Situation

For each episode, participants were asked to indicate the type of situation they were reporting by selecting one of the following categories: work/learning (24%), active leisure (21%), passive leisure (22%), or routine (33%).

8.4 Data Analysis

Because the Psychological Flow Scale has not been validated in an ambulatory assessment setting, we used multilevel CFA to examine the measurement models of the episodic measures, and we tested whether task meaningfulness and the intrinsic reward dimension of flow are distinct constructs. Given the nested data structure (episodic measures within participants), we applied multilevel models with random intercepts and slopes. Models were fit using restricted maximum likelihood. Statistical predictors measured at Level-2 were

grand-mean centered (e.g., personality traits). Predictors measured at Level-1 (e.g., task meaningfulness) were centered in two ways: The person-mean centered variables were included as statistical predictors at Level-1, and the grand-mean centered cluster means were included as statistical predictors at Level-2 (Enders & Tofighi, 2007). This allowed us to test whether and how task meaningfulness was related to flow at the within-person and the between-person level.

We tested our hypotheses as preregistered. Specifically, we tested whether task meaningfulness was positively associated with flow (Model 1), and whether this relation remained robust when skills-demands fit (Model 2) and basic personality traits (Model 3) were included as additional predictors. Further, we explored how task meaningfulness and perceived fit were related to the three flow dimensions absorption (Model 4), effort-less control (Model 5), and intrinsic reward (Model 6). We also explored how task meaningfulness and skills-demands fit were related to flow across different types of situations, to test whether these associations can be generalized to different contexts. In addition, we tested whether the interaction between task meaningfulness and skills-demands fit was associated with flow. Analyses were conducted using R (R Core Team, 2024; version 4.4.1) within RStudio (Posit team, (Posit 2024), 2024.4.2.764), and the following packages: misty (Yanagida, 2024), multilevelTools (Wiley, 2020), lme4 (Bates et al., 2015), and lmerTest (Kuznetsova et al., 2017). To explore the distinctiveness of task meaningfulness and the intrinsic reward dimension of flow, we conducted multilevel CFA using MPLUS (Muthén & Muthén, 1998–2017; version 8.10). Reliability estimates for the episodic measures were obtained as described by Lai (2021), those for the trait measures were obtained as suggested by Flora (2020).

8.5 Transparency and Openness

We report how we determined our sample size, all data exclusions, and all measures relevant to the research question at hand. Data and analysis code, and research materials are available at https://osf.io/d7jrs. This study's design and its analysis were preregistered under https://aspredicted.org/bghk-3b92.pdf.

8.6 Results

The measurement model for flow and task meaningfulness yielded adequate model fit (see Figure S1 in the supplemental material; CFI = 0.97, TLI = 0.96, RMSEA = 0.04, $SRMR_{within}$ = 0.04, $SRMR_{between}$ = 0.05). The loadings of the items on the three flow dimensions (and the loadings on the global flow score) as well as the task meaningfulness factor were positive and significant (see supplemental material). Letting the task meaningfulness items load on the intrinsic reward dimension of flow (i.e., six items for this dimension) resulted in a poor model fit, indicating that task meaningfulness and flow are distinct constructs (CFI = 0.81, TLI = 0.76, RMSEA = 0.09, $SRMR_{within}$ = 0.08, $SRMR_{between}$ = 0.11). The correlations of the separate items are depicted in Table S13 (supplemental material), further supporting this reasoning.

Table 2 contains descriptive statistics, bivariate correlation coefficients, and intraclass correlation coefficients of the involved variables. Both within- and between-person correlations indicated the hypothesized positive relation between task meaningfulness and flow. Further, skills-demands fit was positively associated with flow at within- and between-per-

son level. From the Level-2 variables, Extraversion had the strongest (positive) association with flow. Intraclass correlations of 0.205 and larger imply that more than 20.5% of variability in Level-1 variables could be attributed to inter-individual differences; thus, applying multilevel modeling was an appropriate analysis strategy.

Tables 3 and 4 contain the summary of the multilevel models. Task meaningfulness was positively associated with flow at both within-person and between-person level, supporting our main hypothesis. At the within-person level, Model 1 revealed that, in episodes in which a person reported a value of task meaningfulness that was 1 scale point higher than usual (i.e., the average over all episodes of this person), the person reported a by 0.42 scale points increased flow score. At the between-person level, Model 1 revealed that persons with a value of task meaningfulness that was 1 scale point higher than the average person (i.e., average over all persons in the sample), the person reported a by 0.61 scale points increased flow score. This relation remained robust when skills-demands fit (Model 2) and basic personality traits (Model 3) were included as additional predictors. As illustrated in Fig. 1, skills-demands fit was also positively associated with flow, however, the fixed slope for skills-demands fit was less steep than the one for task meaningfulness. Notably, the 95% confidence interval for the random slopes of task meaningfulness was entirely positive [0.06; 0.70], whereas it included both positive and negative values in the case of skills-demands fit [−0.12; 0.44; refers to Model 3]. Following the approach by Rights and

Table 3 *Multilevel modeling results: predicting flow by Meaningfulness, Skills-Demands Fit, and personality*

Model aspects	Flow								
	Model 1			Model 2			Model 3		
	Est.	SE	Std. Est.	Est.	SE	Sdd. Est.	Est.	SE	Std. Est.
Fixed Effects									
Level 1 – Within									
Task Meaningfulness	0.42	0.01	0.48	0.38	0.01	0.43	0.38	0.01	0.43
Skills-Demands Fit				0.16	0.01	0.17	0.16	0.01	0.17
Level 2 – Between									
Intercept	5.00	0.02		5.00	0.02		5.00	0.02	
Task Meaningfulness	0.61	0.02	0.44	0.47	0.02	0.34	0.45	0.03	0.33
Skills-Demands Fit				0.20	0.02	0.19	0.19	0.02	0.18
Honesty-Humility							0.00	0.02	0.00
Emotionality							0.00	0.02	0.00
Extraversion							0.07	0.02	0.05
Agreeableness							0.04	0.02	0.02
Conscientiousness							0.03	0.02	0.02
Openness							−0.01	0.02	−0.01
Random Effects									
Level 1 – Within	0.54			0.48			0.48		
Level 2 – Between	0.16			0.14			0.13		
Slope for Task Meaningfulness	0.03			0.03			0.03		
Slope for Skills-Demands Fit				0.02			0.02		
Model Summary									
AIC	24,265			23,462			23,493		
BIC	24,316			23,549			23,623		

Note. See Table S15 for exact *p*-values and 95% CIs

Table 4 Multilevel modeling results: predicting Absorption, Effort-less control and intrinsic motivation by Meaningfulness, Skills-Demands Fit, and personality

Model aspects	Absorption			Effort-less control			Intrinsic reward		
	Model 4			Model 5			Model 6		
	Est.	SE	Std. Est.	Est.	SE	Std. Est	Est.	SE	Std. Est
Fixed Effects									
Level 1 – Within									
Task Meaningfulness	0.35	0.01	0.31	0.21	0.01	0.22	0.58	0.01	0.50
Skills-Demands Fit	0.14	0.01	0.12	0.24	0.01	0.24	0.11	0.01	0.09
Level 2 – Between									
Intercept	4.92	0.02		5.32	0.02		4.75	0.02	
Task Meaningfulness	0.47	0.04	0.27	0.28	0.03	0.18	0.60	0.03	0.33
Skills-Demands Fit	0.17	0.03	0.13	0.33	0.02	0.29	0.10	0.02	0.07
Honesty-Humility	0.01	0.03	0.00	0.06	0.02	0.04	−0.08	0.02	−0.05
Emotionality	0.00	0.03	0.00	−0.03	0.03	−0.02	0.04	0.02	0.02
Extraversion	0.03	0.03	0.02	0.09	0.03	0.06	0.07	0.03	0.04
Agreeableness	0.02	0.03	0.01	0.03	0.03	0.02	0.05	0.03	0.02
Conscientiousness	0.06	0.03	0.03	0.03	0.03	0.02	−0.02	0.03	−0.01
Openness	−0.01	0.03	0.00	−0.07	0.02	−0.06	0.05	0.02	0.03
Random Effects									
Level 1 – Within	1.07			0.68			0.95		
Level 2 – Between	0.27			0.24			0.18		
Slope for Task Meaningfulness	0.04			0.03			0.05		
Slope for Skills-Demands Fit	0.03			0.07			0.05		
Model Summary									
AIC	31,636			27,318			30,269		
BIC	31,493			27,448			30,399		

Note. See Table S16 for exact *p*-values and 95% CIs

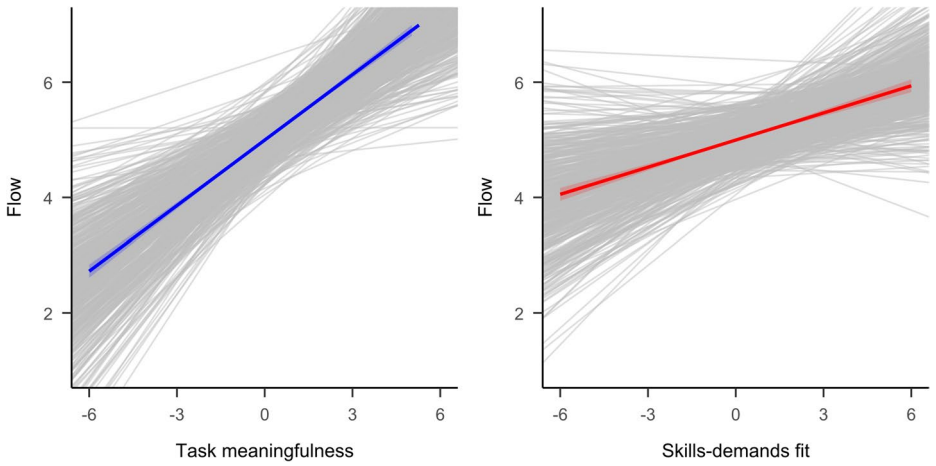


Fig. 1 Fixed and Random Effects of Task Meaningfulness (left panel) and Skills-Demands Fit (right panel) on Flow at the Within-Person Level. Note. At Level-1, task meaningfulness and skills-demands fit were person-mean centered

Sterba (2019), Model 3 explained 62.5% of the variance in flow, whereby Level-1 predictors accounted for 24.8%, Level-2 predictors for 21.9%, random slopes for 5.4%, and random intercepts for 10.4% (see also Figure S2 in the supplemental material).

8.7 Relation between Task Meaningfulness and Flow

In Models 4 through 6, we regressed the dimensions of flow (i.e., absorption, effort-less control, and intrinsic reward) on task meaningfulness, skills-demands fit, and the HEXACO traits. Skills-demands fit and task meaningfulness predicted effort-less control to a comparable extent, but task meaningfulness was more strongly associated with absorption and most strongly related to intrinsic reward. We also examined how task meaningfulness and skills-demands fit were related to flow across different types of situations and found comparable results for work/learning, leisure (active and passive), and routine activities (see supplemental material, Table S14). Descriptively, flow was highest during passive leisure.

Further, we regressed flow on the interaction between skills-demands fit and task meaningfulness and found positive interaction weights at both within-person (Est = 0.01) and between-person level (Est = 0.06; see supplemental material, Table S17). This suggests that skills-demands fit and task meaningfulness were interrelated, such that one aspect was more (less) strongly associated with flow experience when the other aspect was experienced as higher (lower).

8.8 Discussion

Study 2 provides evidence that task meaningfulness and flow experience were positively associated in everyday life. This finding thus provides externally valid support for our proposition. Specifically, multilevel modeling revealed that task meaningfulness was positively related to flow at both within-person and between-person level. This relation remained robust regardless of whether skills-demands fit and basic personality traits were included as additional predictors. In other words, the relation between task meaningfulness and flow was incrementally valid when the most relevant antecedent of flow (i.e., skills-demands fit) and basic personality traits (from the HEXACO model) were controlled for. Moreover, the relation emerged to a comparable extent across different types of situations (e.g., at work and during leisure activities), suggesting that it can be generalized to a wide variety of contexts. It is important to emphasize that we agree with the view that flow cannot occur during passive activities (Barthelmäs & Keller, 2021), because the 'classic' antecedent—skills-demands fit—cannot be met. A passive activity such as watching TV involves no substantive demands with which one's skills could be aligned. Consequently, high scores on the Psychological Flow Scale should be interpreted as evidence of flow only in the context of active tasks.

Because flow experience and both antecedents were assessed simultaneously (shortly after experiencing the episode) in this ambulatory assessment design, the bivariate associations between the constructs were, as expected, higher than in Study 1 (where antecedents and flow were measured separately). The fact that experiencing flow influenced the evaluation of skills-demands fit and task meaningfulness should be considered when interpreting the associations in Study 2. This is particularly relevant for the associations between task meaningfulness and the intrinsic reward dimension of flow. However, it is important to

emphasize that despite the high correlations, the multilevel CFA confirmed that both constructs are distinct. To the best of our knowledge, this is the first study to provide empirical evidence for an adequate model fit of the Psychological Flow Scale in an ambulatory assessment setting.

The results also indicate that skills-demands fit and task meaningfulness differed in how they were specifically related to flow, revealing a pattern similar to that observed in Study 1. Task meaningfulness was particularly associated with intrinsic reward and absorption, whereas both antecedents showed comparable associations with effort-less control. We found an interaction between task meaningfulness and skills-demands fit in predicting flow. Specifically, when one aspect was lower (or higher), the other aspect was also less (or more) strongly associated with flow experience. This suggests that experiencing flow may require a certain level of both task meaningfulness and skills-demands fit, as the absence of one may not be fully compensated by the other. This finding aligns with the pattern observed in Study 1, which indicated that both antecedents operate independently but are likely not entirely unrelated.

9 General Discussion

The present work aimed to test the proposition that task meaningfulness is a relevant antecedent of flow experience. Although similar ideas have been proposed in the literature, they have neither been embedded in a broader theoretical framework nor empirically tested. We tested this proposition in a controlled experimental setting and in an ambulatory assessment study (both preregistered), providing internally and externally valid evidence from more than 1,000 participants and more than 10,000 episodic measures that task meaningfulness is a relevant antecedent of flow experience.

The studies suggest that both antecedents show differential associations with experiential aspects of flow. Task meaningfulness was stronger related to the intrinsic reward and absorption dimensions of flow than to the effort-less control dimension, both in the experimental and the ambulatory assessment setting. The theoretical underpinning of our main proposition is well in line with this interpretation. Based on the JCM, it is plausible to expect that task meaningfulness is specifically related to the intrinsic motivation aspect of experience. It is also plausible that the effort-less control dimension of flow is a function of skills-demands fit. The extent to which task demands are in balance with the individual's skills most directly influences the perception of how easy it is to complete a task. Interestingly, the variability of the random slopes indicated that the relation between skills-demands fit and flow was negative (or zero) for about 25% of the participants, whereas the relation between task meaningfulness and flow was positive for almost all participants. This finding is well in line with former work demonstrating that the relation between skills-demands fit and flow is substantially qualified by additional factors (Baumann et al., 2016; Engeser & Baumann, 2016; Keller & Bless, 2008; Keller & Blomann, 2008). The positive relation between task meaningfulness and flow could thus be considered as pretty robust for many individuals and situations, although the relation between skills-demands fit and flow seems to be qualified by characteristics of the individual and the situation. Acknowledging that we did not consider moderators for the relation between skills-demands fit and flow in the present study may account for why the effect size of this relation was smaller than the rela-

tion between task meaningfulness and flow. We suggest avoiding any overemphasis on the larger regression weight for task meaningfulness compared to the one for skills-demands fit.

We acknowledge that our results do not definitively resolve the question of whether the antecedents are fully independent or not. While Study 1 did not reveal a significant interaction term between the two antecedents in the regression analyses, we did observe a small descriptive effect of the manipulation on skills-demands fit. Combined with the interaction effect found in Study 2, the most reasonable conclusion is that task meaningfulness and skills-demands fit are largely, but not entirely, independent antecedents of flow. Experiencing flow may require at least a minimum level of both task meaningfulness and skills-demands fit, as the absence of one may not be fully compensated by the other. Future studies could explore this further, for example by systematically varying both task meaningfulness and skills-demand fit simultaneously. This approach could also clarify whether these two factors contribute to the experience of flow through distinct or similar underlying processes.

9.1 Distinguishing Task Meaningfulness and Outcome Importance

In this context, it is also important to distinguish task meaningfulness from outcome importance, a conceptually different construct that has previously been shown to interact with skills-demands fit in predicting flow experiences. Engeser and Rheinberg (2008) found support for their hypothesis that the outcome importance of an activity moderates the relation between skills-demands fit and flow. Specifically, activities with high outcome importance were associated with high flow scores when skills exceeded demands, whereas activities with low outcome importance showed the highest flow when skills and demands were balanced. Similarly, Engeser and Baumann (2016) demonstrated that considering outcome importance can partially explain differences in flow levels between work and leisure. However, it is important to clarify that outcome importance is not the same construct as task meaningfulness. This distinction becomes evident when examining the respective items. Specifically, items assessing outcome importance primarily capture aspects of fear of failure (e.g., 'I am worried about failing'). Empirically, momentary outcome importance has been shown to be negatively associated with momentary happiness/satisfaction (Engeser & Baumann, 2014). In contrast, for task meaningfulness, one would expect the opposite pattern—a positive association with happiness and satisfaction (Arnold et al., 2007; Steger et al., 2012). Outcome importance and task meaningfulness are therefore not synonymous but distinct constructs that should be considered separately.

9.2 Task Meaningfulness and Flow Experience Can Be Considered as Distinct Constructs

Three considerations are particularly relevant to the question of whether task meaningfulness and the intrinsic reward dimension of flow are distinct constructs. First, we found a poor model fit in both studies if the three items used to measure task meaningfulness were fixed to load onto the flow dimension intrinsic reward. These empirical results show that both constructs are distinct. Second, several theoretical approaches clearly separate the task meaningfulness and intrinsic motivational aspects of experience. This applies to the aforementioned JCM (Hackman & Oldham, 1976) and to SDT (Ryan & Deci, 2017). Westgate and Wilson's (2018; see also Westgate, 2020) Meaning and Attentional Components

(MAC) model takes a similar perspective. They proposed that the state of boredom, which represents an opposite pole to flow, is the result of deficits in attention (i.e., a mismatch between cognitive demands and available mental resources) and/or meaning (i.e., mismatches between activities and valued goals). The MAC model also makes a clear distinction between the antecedent (i.e., meaning) and the state (i.e., boredom). Third, no known theoretical or empirical work on the flow experience has conceptualized task meaningfulness as an experiential component of the flow experience. In summary, both theoretical and empirical arguments indicate that task meaningfulness and intrinsic reward are separable constructs, thereby mitigating concerns of circular reasoning in our theoretical rationale, empirical evidence, and conclusions.

9.3 Future Directions

As previously mentioned, it would be worthwhile to develop study designs in which both task meaningfulness and skills-demands fit can be manipulated simultaneously. Such designs could help clarify whether a minimum level of both antecedents is necessary for the emergence of flow, and whether these two factors interact with one another. For example, the meta-analysis by Steingut et al. (2017) found that rationale provision was especially conducive to engagement and performance in rather boring tasks, suggesting that task meaningfulness and skills-demands fit may be non-linearly interdependent. In Study 1, our manipulation targeted the 'greater goods motivation' component of task meaningfulness by linking Tetris performance to a charitable donation. Applying a rationale to increase task meaningfulness, however, seems less feasible in the context of Tetris—at least, we could not identify a compelling argument to support such an approach. In contrast, activities such as learning or sports may be better suited for this purpose. While these settings offer less experimental control, they allow for the investigation of complex tasks such as skill acquisition, which may provide more opportunities to leverage the effects of rationale provision. Future studies could also explore boundary conditions under which rationale provision might produce backfiring effects. For instance, emphasizing the importance of basic arithmetic skills might trigger (stereotype) threat in certain individuals, thereby undermining their motivation.

9.4 Limitations

We measured skills-demands fit with a single item in both studies. Future studies should assess this antecedent of flow as nuanced as task meaningfulness to demonstrate that our results are not a function of measurement precision. Although our samples are more diverse than typical student samples and offer valuable insights, they remain limited due to their nonrepresentative nature and exclusive focus on the German population. As in these studies, flow is typically conceptualized as a continuously distributed construct. However, there are also theoretical considerations suggesting that flow may be treated as a binary phenomenon (e.g., Peifer & Engeser, 2021). If future research develops a valid and reliable approach for conceptualizing flow categorically, it will be interesting to examine whether the present findings can be replicated under such a framework.

10 Conclusion

The present paper provides empirical evidence from both a controlled experimental setting and from everyday life that task meaningfulness is a relevant antecedent of flow experience. Both studies suggest that task meaningfulness and skills-demand fit (the most relevant antecedent of flow to date) show differential associations with experiential aspects of flow and might promote flow through different processes. Accordingly, addressing task meaningfulness may provide the basis for innovative and effective flow interventions.

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Author Contributions **Michael Barthelmäs**: Conceptualization, Methodology, Formal analysis, Resources, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration, Funding acquisition. **Dominik Stöckle**: Methodology, Investigation, Writing - Original Draft. **Johannes Keller**: Conceptualization, Writing - Review & Editing.

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Data Availability Data, analysis code, and research materials are available at https://osf.io/d7jrs.

Declarations

Conflict of interest The authors report there are no competing interests to declare.

Ethical Approval Approval was obtained from the local ethics committee (No. 410/19, 13.02.2020). The procedures used in these studies adhere to the tenets of the Declaration of Helsinki.

Consent To Participate Informed consent was obtained from all individual participants included in the studies.

Consent To Publish All participants signed informed consent regarding publishing their data.

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